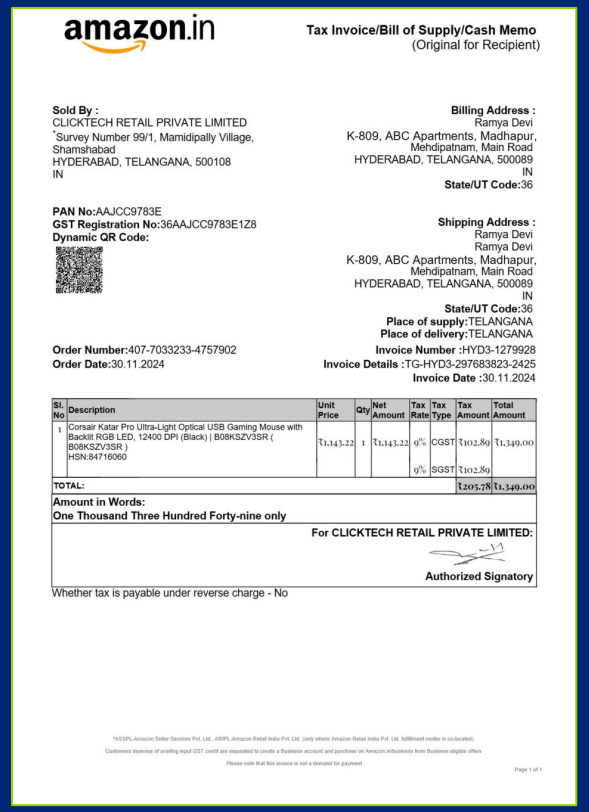
Project

OpenAI API | GPT | Vision Models | MongoDB | LangChain | FastAPI

Intelligent Document Processing with OpenAI GPT & LangChain

OpenAI API | GPT | Vision Models | MongoDB | LangChain | FastAPI

Project



amazon.in

Tax Invoice/Bill of Supply/Cash Memo
(Original for Recipient)

Sold By :
CLICKTECH RETAIL PRIVATE LIMITED
Survey Number 99/1, Mamidipally Village, Shamshabad
HYDERABAD, TELANGANA, 500108
IN

PAN No:AAJCC9783E
GST Registration No:36AAJCC9783E1Z8
Dynamic QR Code:

Order Number:407-7033233-4757902
Order Date:30.11.2024

Billing Address :
Ramya Devi
K-809, ABC Apartments, Madhapur, Mehdipatnam, Main Road
HYDERABAD, TELANGANA, 500089
IN
State/UT Code:36

Shipping Address :
Ramya Devi
Ramya Devi
K-809, ABC Apartments, Madhapur, Mehdipatnam, Main Road
HYDERABAD, TELANGANA, 500089
IN
State/UT Code:36
Place of supply:TELANGANA
Place of delivery:TELANGANA
Invoice Number :HYD3-1279928
Invoice Details :TG-HYD3-297683823-2425
Invoice Date :30.11.2024

Sl. No	Description	Unit Price	Qty	Net Amount	Tax Rate	Tax Type	Tax Amount	Total Amount
1	Corsair Katar Pro Ultra-Light Optical USB Gaming Mouse with Backlit RGB LED, 12400 DPI (Black) | B08KSZV3SR (B08KSZV3SR) HSN:84716060	₹1,143.22	1	₹1,143.22	9%	CGST	₹102.80	₹1,349.00
					9%	SGST	₹102.80	
TOTAL:							₹205.78	₹1,349.00

Amount in Words:
One Thousand Three Hundred Forty-nine only

For CLICKTECH RETAIL PRIVATE LIMITED:

Authorized Signatory

Whether tax is payable under reverse charge - No

*ASSPL-Amazon Seller Services Pvt. Ltd., ARIPL-Amazon Retail India Pvt. Ltd. (only where Amazon Retail India Pvt. Ltd. fulfillment center is co-located)

Customers desirous of availing input GST credit are requested to create a Business account and purchase on Amazon.in/business from Business eligible offers

Please note that this invoice is not a demand for payment

Page 1 of 1

ACE
The helpful place.

Thank You for Shopping at
Ace Hardware of Navarre
8188 Navarre Pkwy, Navarre, FL 32566
Phone: (850) 939-9800

11/14/2024, 5:46:31 PM

Reg: 2 ACE Rewards # : 1928777
Customer: J Manchester: Since 2011

2489540 DeWalt 20V MAX Cordless Brushed 2 Tool Kit $219.99
2029735 DeWalt Maxfit Screwdriver Bit Set 30 pc $19.98
CLDBVG COKE 20oz $1.99

Subtotal $241.97
Tax $16.93

Total $258.90

Card number **** **** **** 3081
Card type Debit
Card entry Chip
Date/time 11/14/2024 5:46 PM
Reference # 628452892602462406 85C
Status APPROVED

THANK YOU
HAVE A NICE DAY

RAFAEL ULATE
DIRECTOR OF ADMISSIONS
DEPARTMENT OF ELECTRICAL ENGINEERING

DAVID PACKARD ELECTRICAL ENGINEERING
350 SERRA MALL, ROOM 170
STANFORD UNIVERSITY
STANFORD, CALIFORNIA 94305-9505

TEL: (650) 725-9327
FAX: (650) 723- 1882
ulate@ee.stanford.edu

भारत सरकार
GOVERNMENT OF INDIA

NAME : XXXX
DOB : XXXX
GENDER : XXXX

4444 3333 6666 8888

आधार - भारतीय विशिष्ट पहचान प्राधिकरण

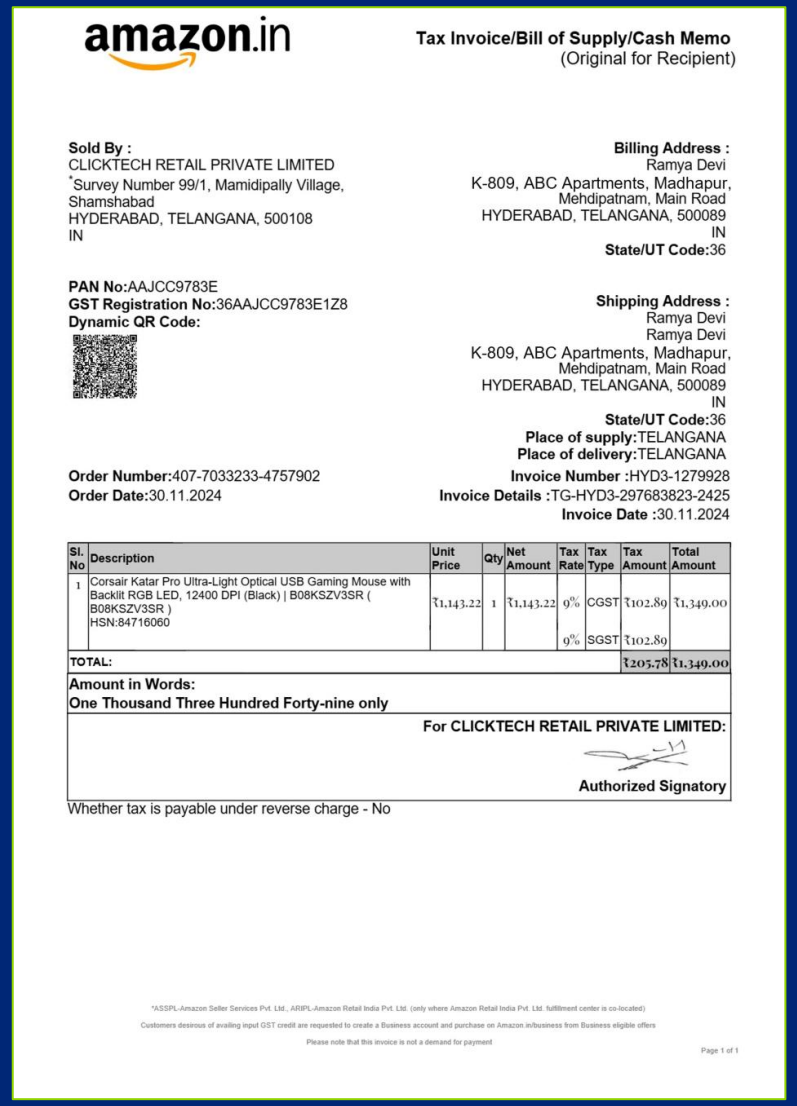
SBI भारतीय स्टेट बैंक
STATE BANK OF INDIA

पासबुक
PASS BOOK

किओस्क बैंक
KIOSK BANK

vedicskill.

Invoice



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GST Registration No:36AAJCC9783E1Z8
Dynamic QR Code:

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HYDERABAD, TELANGANA, 500089
IN
State/UT Code:36

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Ramya Devi
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HYDERABAD, TELANGANA, 500089
IN
State/UT Code:36
Place of supply:TELANGANA
Place of delivery:TELANGANA

Order Number:407-7033233-4757902
Order Date:30.11.2024

Invoice Number :HYD3-1279928
Invoice Details :TG-HYD3-297683823-2425
Invoice Date :30.11.2024

Sl. No	Description	Unit Price	Qty	Net Amount	Tax Rate	Tax Type	Tax Amount	Total Amount
1	Corsair Katar Pro Ultra-Light Optical USB Gaming Mouse with Backlit RGB LED, 12400 DPI (Black) | B08KSZV3SR (B08KSZV3SR) HSN:84716060	₹1,143.22	1	₹1,143.22	9%	CGST	₹102.89	₹1,349.00
					9%	SGST	₹102.89	
TOTAL:							₹205.78	₹1,349.00

Amount in Words:
One Thousand Three Hundred Forty-nine only

For CLICKTECH RETAIL PRIVATE LIMITED:

Authorized Signatory

Whether tax is payable under reverse charge - No

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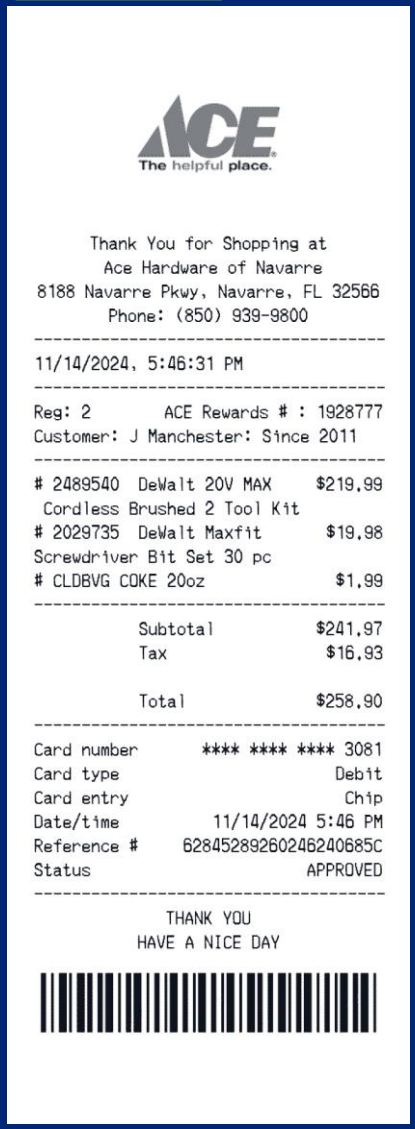
Receipt



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Reg: 2 ACE Rewards # : 1928777
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2489540 DeWalt 20V MAX $219.99
Cordless Brushed 2 Tool Kit
2029735 DeWalt Maxfit $19.98
Screwdriver Bit Set 30 pc
CLDBVG COKE 20oz $1.99

Subtotal $241.97
Tax $16.93

Total $258.90

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HAVE A NICE DAY

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TEL: (650) 725-9327
FAX: (650) 723- 1882
ulate@ee.stanford.edu

Business Card

भारत सरकार
GOVERNMENT OF INDIA

NAME : XXXX
DOB : XXXX
GENDER : XXXX

4444 3333 6666 8888

आधार - भारतीय विशिष्ट पहचान प्राधिकरण

Government ID Cards

SBI भारतीय स्टेट बैंक
STATE BANK OF INDIA

पासबुक
PASS BOOK

yono SBI

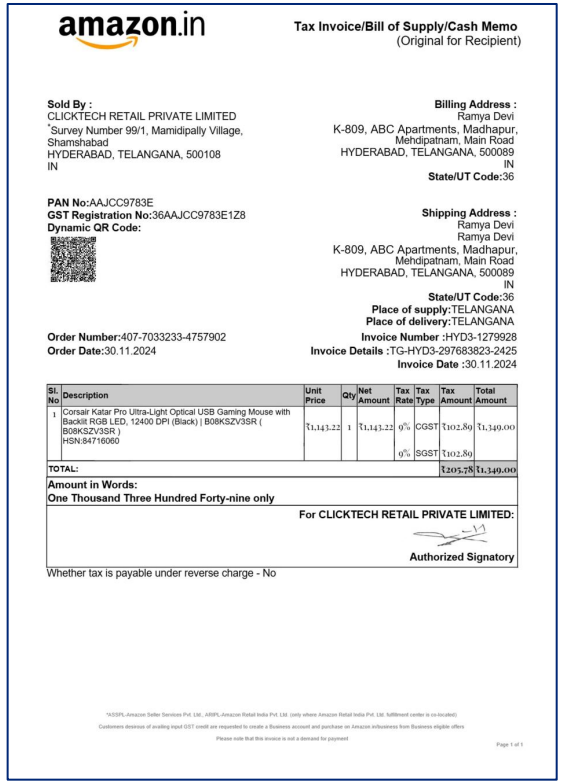
किओस्क बैंक
KIOSK BANK

Bank Documents

Documents

In this course we will focus on two documents

Invoice



amazon.in

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CLICKTECH RETAIL PRIVATE LIMITED
*Survey Number 99/1, Mamidipally Village, Shamshabad
HYDERABAD, TELANGANA, 500108
IN

PAN No:AAJCC9783E
GST Registration No:36AAJCC9783E1Z8
Dynamic QR Code:

Order Number:407-7033233-4757902
Order Date:30.11.2024

Billing Address :
Ramya Devi
K-809, ABC Apartments, Madhapur, Mehdipatnam, Main Road
HYDERABAD, TELANGANA, 500089
IN
State/UT Code:36

Shipping Address :
Ramya Devi
Ramya Devi
K-809, ABC Apartments, Madhapur, Mehdipatnam, Main Road
HYDERABAD, TELANGANA, 500089
IN
State/UT Code:36
Place of supply:TELANGANA
Place of delivery:TELANGANA

Invoice Number :HYD3-1279928
Invoice Details :TG-HYD3-297683823-2425
Invoice Date :30.11.2024

Sl. No	Description	Unit Price	Qty	Net Amount	Tax Rate	Tax Type	Tax Amount	Total Amount
1	Corsair Katar Pro Ultra-Light Optical USB Gaming Mouse with Backlit RGB LED, 12400 DPI (Black) | B08KSZV3SR (B08KSZV3SR) HSN:84716060	₹1,143.22	1	₹1,143.22	9%	CGST	₹102.89	₹1,349.00
					9%	SGST	₹102.89	
TOTAL:							₹205.78	₹1,349.00

Amount in Words:
One Thousand Three Hundred Forty-nine only

For CLICKTECH RETAIL PRIVATE LIMITED:

Authorized Signatory

Whether tax is payable under reverse charge - No

*KSSPL-Amazon Seller Services Pvt. Ltd., ARIPL-Amazon Retail India Pvt. Ltd. (only where Amazon Retail India Pvt. Ltd. fulfillment center is co-located)

Customers desirous of availing input GST credit are requested to create a Business account and purchase on Amazon inBusiness from Business eligible offers

Please note that this invoice is not a demand for payment

Page 1 of 1

Business Card

Seller Name

Seller Address

Buyer Name

Buyer Name

Invoice Number

Invoice Date

Items / Description

Total Amount

Currency

amazon.in

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*Survey Number 99/1, Mamidipally Village, Shamshabad
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Mehdipatnam, Main Road
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State/UT Code:36

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Ramya Devi
K-809, ABC Apartments, Madhapur,
Mehdipatnam, Main Road
HYDERABAD, TELANGANA, 500089
IN
State/UT Code:36
Place of supply:TELANGANA
Place of delivery:TELANGANA

Order Number:407-7033233-4757902
Order Date:30.11.2024

Invoice Number :HYD3-127992
Invoice Details :TG-HYD3-297683823-2425
Invoice Date :30.11.2024

Sl. No	Description	Unit Price	Qty	Net Amount	Tax Rate	Tax Type	Tax Amount	Total Amount
1	Corsair Katar Pro Ultra-Light Optical USB Gaming Mouse with Backlit RGB LED, 12400 DPI (Black) | B08KSZV3SR (B08KSZV3SR) HSN:84716060	₹1,143.22	1	₹1,143.22	9%	CGST	₹102.89	₹1,349.00
					9%	SGST	₹102.89	
TOTAL:							₹205.78	₹1,349.00

Amount in Words:
One Thousand Three Hundred Forty-nine only

For CLICKTECH RETAIL PRIVATE LIMITED:

Authorized Signatory

Whether tax is payable under reverse charge - No

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Page 1 of 1

amazon.in

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Sold By :
CLICKTECH RETAIL PRIVATE LIMITED
*Survey Number 99/1, Mamidipally Village, Shamshabad
HYDERABAD, TELANGANA, 500108
IN

PAN No:AAJCC9783E
GST Registration No:36AAJCC9783E1Z8
Dynamic QR Code:

Billing Address :
Ramya Devi
K-809, ABC Apartments, Madhapur, Mehdipatnam, Main Road
HYDERABAD, TELANGANA, 500089
IN
State/UT Code:36

Shipping Address :
Ramya Devi
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K-809, ABC Apartments, Madhapur, Mehdipatnam, Main Road
HYDERABAD, TELANGANA, 500089
IN
State/UT Code:36
Place of supply:TELANGANA
Place of delivery:TELANGANA

Order Number:407-7033233-4757902
Order Date:30.11.2024

Invoice Number :HYD3-1279928
Invoice Details :TG-HYD3-297683823-2425
Invoice Date :30.11.2024

Sl. No	Description	Unit Price	Qty	Net Amount	Tax Rate	Tax Type	Tax Amount	Total Amount
1	Corsair Katar Pro Ultra-Light Optical USB Gaming Mouse with Backlit RGB LED, 12400 DPI (Black) | B08KSZV3SR (B08KSZV3SR) HSN:84716060	₹1,143.22	1	₹1,143.22	9%	CGST	₹102.89	₹1,349.00
					9%	SGST	₹102.89	
TOTAL:							₹205.78	₹1,349.00

Amount in Words:
One Thousand Three Hundred Forty-nine only

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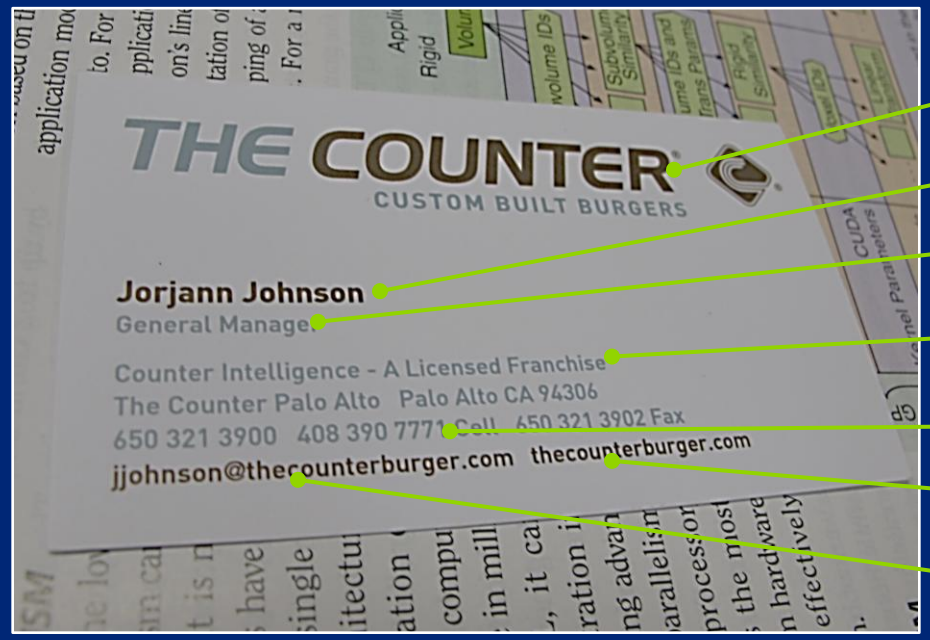
Page 1 of 1

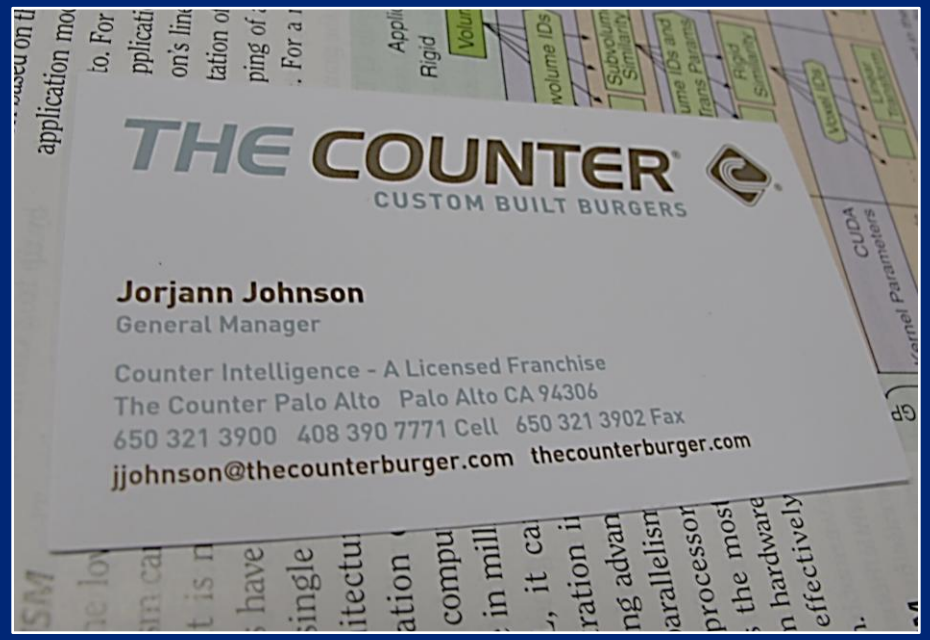
Output from OpenAI GPT Model


```
{
  "invoice_number": "HYD3-1279928",
  "invoice_date": "30.11.2024“
  "seller_name": "CLICKTECH RETAIL PRIVATE LIMITED“
  "buyer_name": "Ramya Devi",
  "seller_address": [
                    "SURVEY NUMBER 99/1, Mamidipally Village,
                     Shamshabad, HYDERABAD, TELANGANA,
                     500108,IN“
          ],
, "buyer_address": [
                    "K-809, ABC Apartments, Madhapur,
                     Mehdipatnam, Main Road, HYDERABAD,
                     TELANGANA, 500089, IN“
          ],
  "total_amount": "1349.00",
  "currency": "INR",
  "items": [
            {
              "item_number": "1",
              "description": [
            "Corsair Katar Pro Ultra-Light Optical USB
             Gaming Mouse with Backlit RGB LED, 12400 DPI
             (Black)“
          ],
   "quantity": "1",
   "item_amount": "1349.00“
          }
    ]
}
```



Business Card

- Company
- Name
- Designation/Title
- Address
- Phone Numbers
- Website
- Email



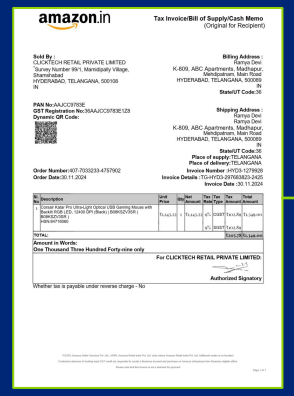
Business Card


```
{
  "name": "Jorjann Johnson",
  "title": "General Manager",
  "company": "The Counter",
  "phone_numbers": [
      "650-321-3900",
      "408-390-7771",
      "650-321-3902"
   ],
 "emails": [
      "jjohnson@thecounterburger.com"
   ],
 "website": "thecounterburger.com“,
 "address": [
      "The Counter, Palo Alto, CA 94306“
   ],
}
```

Architecture

Intelligent Document Extraction with OpenAI GPT & LangChain

→ Python, OpenAI API, Langchain, MongoDB


amazon.in

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IN
State/UT Code:36
Place of supply:TELANGANA
Place of delivery:TELANGANA

Order Number:407-7033233-4757902
Order Date:30.11.2024

Invoice Number :HYD5-1279926
Invoice Details :TG-HYD5-287683823-2425
Invoice Date :30.11.2024

No.	Description	Unit Price	Qty	Net Amount	Tax Rate	Tax Type	Tax Amount	Total Amount
1	Zunight Kaler Pro Ultra-Light Optical USB Gaming Mouse with Sharkk RGB LED, 12400 DPI (Black) | B08K52VR9R | ASIN:B07X6BC | ASIN:B07X6BC	₹1,249.15	1	₹1,249.15	5%	GSTIN	₹49.96	₹1,249.15
					5%	IGSTIN	₹49.96	
TOTAL:							₹99.92	₹1,249.15

Amount in Words:
One Thousand Three Hundred Forty-nine only

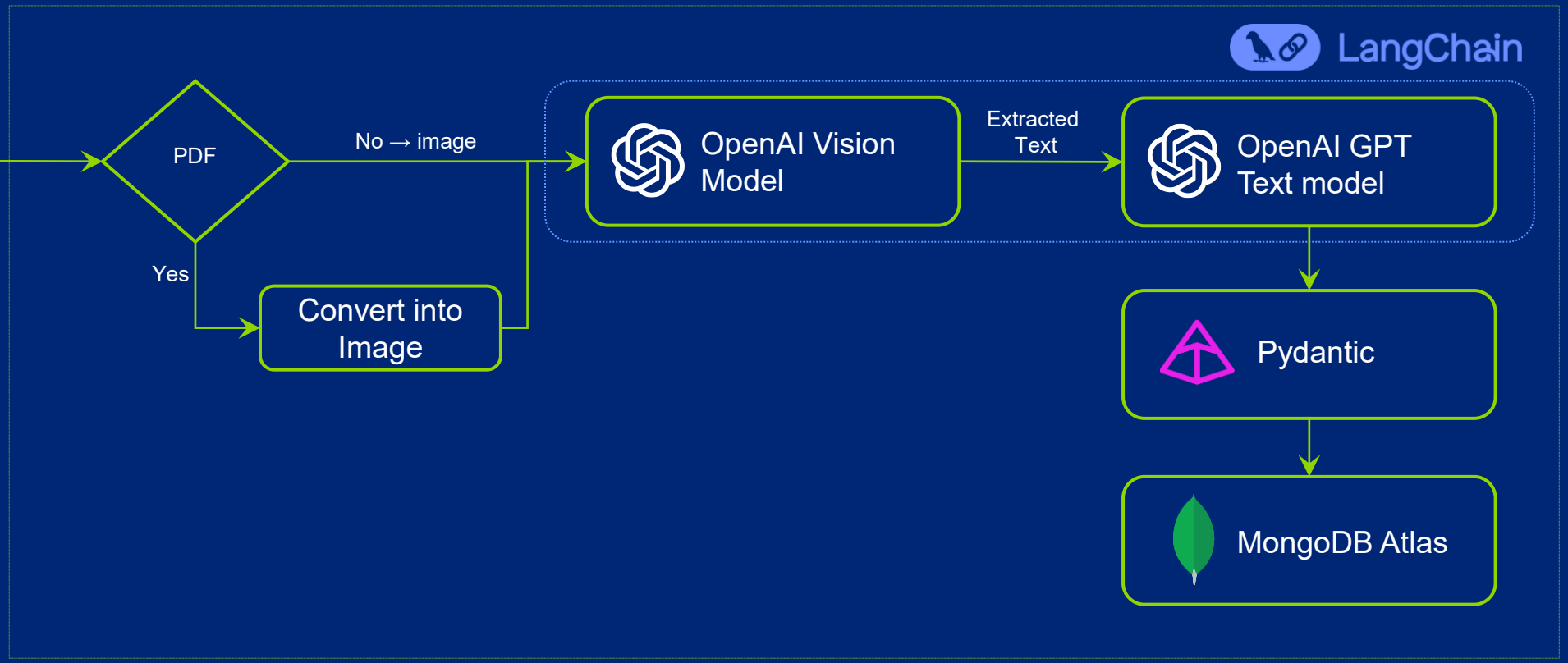
For CLICKTECH RETAIL PRIVATE LIMITED:

Authorized Signatory

Whether tax is payable under reverse charge - No

"AGORA Amazon Data Services Pvt. Ltd., AMZL Amazon Retail India Pvt. Ltd. (also stores Amazon Retail India Pvt. Ltd. fulfillment centers in Amazon) Customer service of sending back GST credit requested results Amazon account and purchase on Amazon.in/amazon.from Amazon orders. Also Please note that this invoice is not a national tax payment

Page 1 of 1



vedicskill.

Next Development

Folder Structure

Create the folder structure for our project

Folder Structure

```
document_ai/
│
├── main_notebook.ipynb
│
├── config/
│   └── settings.py
│
├── models/
│   └── invoice_model.py
│
├── prompts/
│   └── invoice_prompt.py
│
├── services/
│   ├── file_handler.py
│   ├── vision_service.py
│   ├── extraction_service.py
│   ├── validation_service.py
│   └── storage_service.py
│
└── db/
    └── mongodb.py
```

File handling

Handling image and pdf

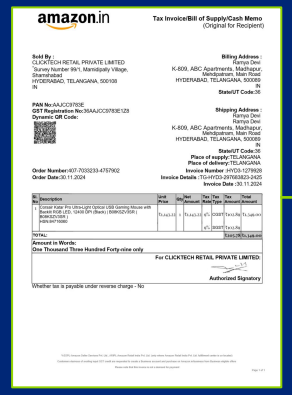
File Handling

`file_handler.py`

Two type of documents

→ **PDF**

→ **Image (PNG, JPG, JPEG)**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Mindspacesm, Main Road
HYDERABAD, TELANGANA, 500089
IN
State/UT Code:36
Place of supply:TELANGANA
Place of delivery:TELANGANA

Order Number:407-7032233-4757902
Order Date:30.11.2024

Invoice Number :HYD3-1279828
Invoice Details :TD-HYD3-2876838223-2425
Invoice Date :30.11.2024

Sl. No.	Description	Unit Price	Qty	Net Amount	Tax Rate	Tax Type	Tax Amount	Total Amount
1	Corsair Katar Pro Ultra-Light Optical USB Gaming Mouse with Black8 RGB LED, 12400 DPI (Black) (B08K6ZVBGR) Product Code: katar-pro-black-usb-eu-nl-no-fi-dk-se-pl-tr-uae-sa-eg-za-ae

File Handling

Dependency libraries

Handle PDF

- PyMuPDF

```
pip install pymupdf
      or
poetry add pymupdf
```

Handle Image

- PIL

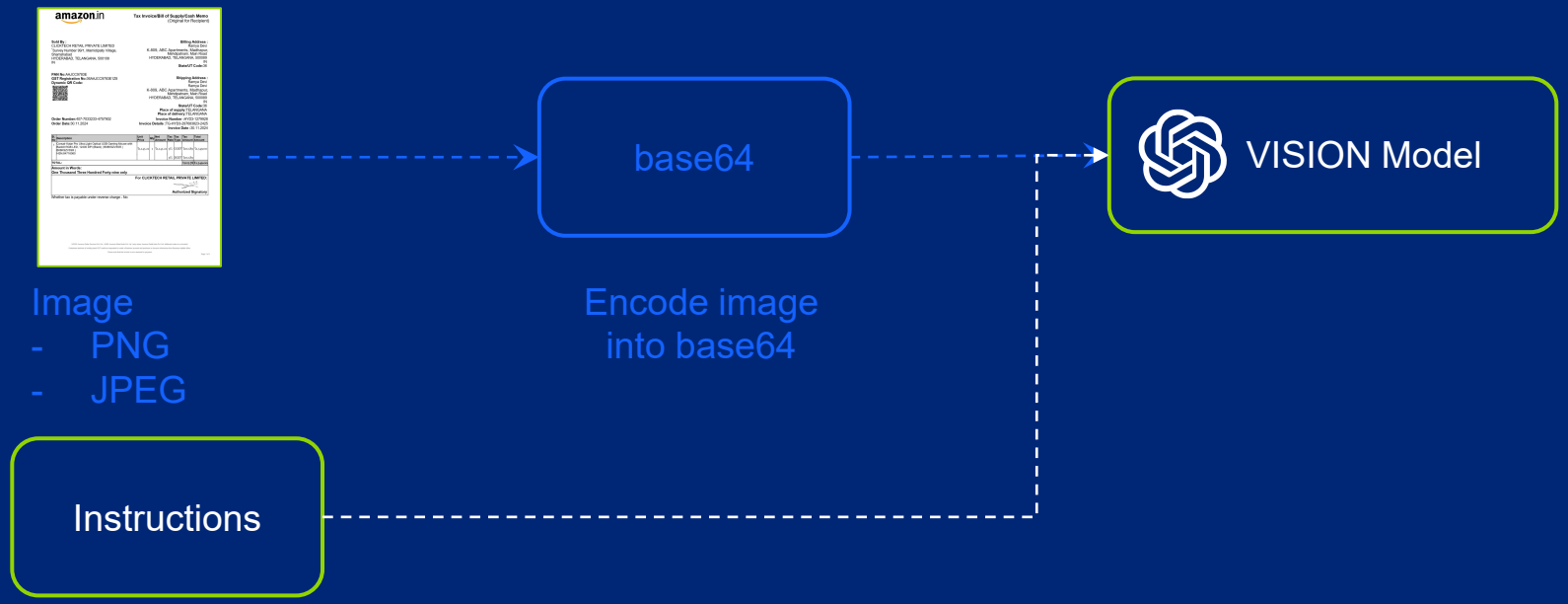
Vision Service

Build LLM framework handle image as input

Vision Model

`vision_service.py`

→ Build function for vision model



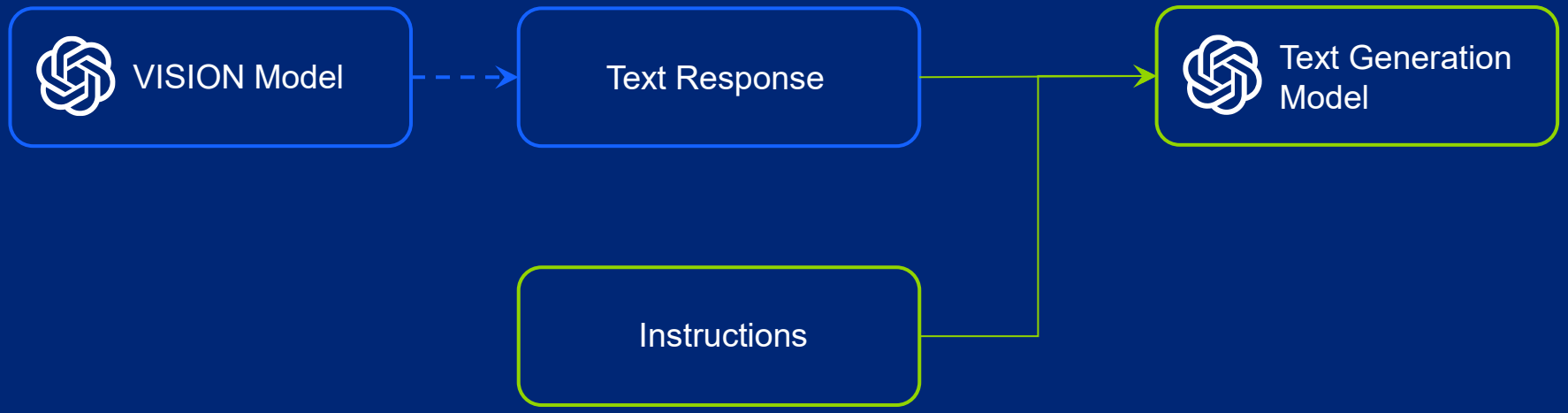
Extraction Service

Create Prompt to get entities from the text

Extraction Model

`extraction_serivce.py`

→ Extract entitles from the extracted text



Validation Service

Create Prompt to get entities from the text

Pydantic Data Validation

LLM Output

```
data = {
 "invoice_number": "422160369",
 "buyer": "ATTN: Data Science Anywhere",
 "seller": "Amazon Internet Services Private Ltd.",
 "invoice_date": "2020-04-03",
 "items": [
            {
                        "description": "Charges",
                        "quantity": "",
                        "price": "764.30",
                        "total": "764.30"
            },
            {
                        "description": "GST",
                        "quantity": "",
                        "price": "137.94",
                        "total": "137.94"
            }
           ],
 "subtotal": "764.30",
 "tax": "137.94",
 "total": "902.24“
}
```

Pydantic Model Dump

```
data = {
 'invoice_number': '422160369’,
 'buyer': 'ATTN: Data Science Anywhere’,
 'seller': 'Amazon Internet Services Private Ltd.’,
 'invoice_date': datetime.date(2020, 4, 3),
 'items’: [
            {
                        'description': 'Charges’,
                        'quantity’: None,
                        'price': 764.3,
                        'total': 764.3
            },
            {
                        'description': 'GST’,
                        'quantity': None,
                        'price': 137.94,
                        'total': 137.94
            }
            ],
 'subtotal': 764.3,
 'tax': 137.94,
 ‘total’: 902.24
}
```

Pydantic Data Validation

```
data = {
 'invoice_number': '422160369’,
 'buyer': 'ATTN: Data Science Anywhere’,
 'seller': 'Amazon Internet Services Private Ltd.’,
 'invoice_date': datetime.date(2020, 4, 3),
 'items’: [
          {
                    'description': 'Charges’,
                    'quantity’: None,
                    'price': 764.3,
                    'total': 764.3
          },
          {
                   'description': 'GST’,
                   'quantity': None,
                   'price': 137.94,
                   'total': 137.94
          }
          ],
 'subtotal': 764.3,
 'tax': 137.94,
 ‘total’: 902.24
}
```

```
BaseModel → Item

description: str
quantity: int
price: float
total: float
```

```
BaseModel → Invoice

invoice_number: str
buyer: str
seller: str
invoice_date: datetime,
items: List[Item]
subtotal: float,
tax: float,
total: float
```

```
Model Validator

→ Replace empty string with None
```

Pydantic Data Models

```
BaseModel → Item

description: str
quantity: int
price: float
total: float
```

```
BaseModel → Invoice

invoice_number: str
buyer: str
seller: str
invoice_date: datetime,
items: List[Item]
subtotal: float,
tax: float,
total: float
```

```
Model Validator

→ Replace empty string with None
```

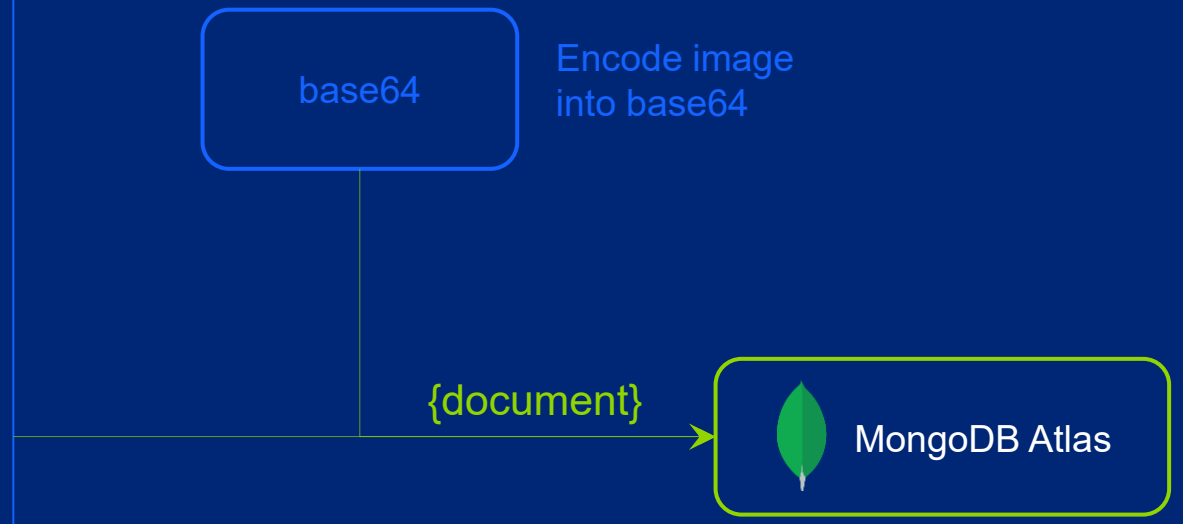
Storage Service

Store data in MongoDB

Storage Service

Save invoice output data also with base64 image in MongoDB

```
data = {
 'invoice_number': '422160369’,
 'buyer': 'ATTN: Data Science Anywhere’,
 'seller': 'Amazon Internet Services Private Ltd.’,
 'invoice_date': datetime.date(2020, 4, 3),
 'items’: [
          {
                     'description': 'Charges’,
                     'quantity’: None,
                     'price': 764.3,
                     'total': 764.3
          },
          {
                    'description': 'GST’,
                    'quantity': None,
                    'price': 137.94,
                    'total': 137.94
          }
          ],
 'subtotal': 764.3,
 'tax': 137.94,
 ‘total’: 902.24
}
```



Next

Document AI web App

Create Web App in FastAPI and Integrate all services